DSA0405-Fundamentals of Data Science - Lab Manual(1-6)

1. Average score for each subject and highest average

```
import numpy as np
student_scores = np.array([
    [80, 75, 85, 70],
    [90, 80, 78, 85],
    [70, 85, 80, 75],
    [85, 90, 88, 80]
])
subjects = ["Math", "Science", "English", "History"]
# Calculate average of each subject
average_scores = np.mean(student_scores, axis=0)
# Find subject with highest average
highest_subject = subjects[np.argmax(average_scores)]
print("Average scores:", average_scores)
print("Subject with highest average:", highest_subject)
```

Output:

```
import numpy as np

student_scores = np.array([
    [80, 75, 85, 70],
    [90, 80, 78, 85],
    [70, 85, 80, 75],
    [85, 90, 88, 80]
])

subjects = ["Math", "Science", "English", "History"]

# Calculate average of each subject
average_scores = np.mean(student_scores, axis=0)

# Find subject with highest average
highest_subject = subjects[np.argmax(average_scores)]

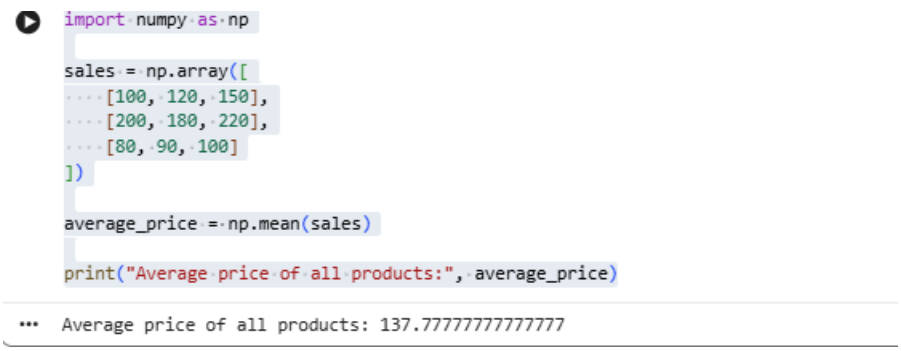
print("Average scores:", average_scores)
print("Subject with highest average:", highest_subject)
```

```
Average scores: [81.25 82.5  82.75 77.5 ]
Subject with highest average: English
```

2. Average price of all products sold

```
import numpy as np
sales = np.array([
    [100, 120, 150],
    [200, 180, 220],
    [80, 90, 100]
])
average_price = np.mean(sales)
print("Average price of all products:", average_price)
```

Output:



3. Average sale price of houses with more than four bedrooms

```
import numpy as np
house_data = np.array([
    [3, 1500, 250000],
    [5, 2200, 400000],
    [4, 1800, 300000],
    [6, 2500, 500000],
    [5, 2100, 450000]
])
filtered_houses = house_data[house_data[:, 0] > 4]
```

average_price = np.mean(filtered_houses[:, 2])

print("Average sale price:", average_price)

Output:

```
# 3. Average sale price of houses with more than four bedrooms

import numpy as np

house_data = np.array([
    [3, 1500, 250000],
    [5, 2200, 400000],
    [4, 1800, 300000],
    [6, 2500, 500000],
    [5, 2100, 450000]
])

filtered_houses = house_data[house_data[:, 0] > 4]
average_price = np.mean(filtered_houses[:, 2])

print("Average sale price:", average_price)
```

```
Average sale price: 450000.0
```

4. Total sales and percentage increase

import numpy as np

sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)

percentage_increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100

print("Total sales:", total_sales)

print("Percentage increase:", percentage_increase, "%")

Output:

```
# 4. Total sales and percentage increase

import numpy as np

sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)
percentage_increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100

print("Total sales:", total_sales)
print("Percentage increase:", percentage_increase, "%")
```

```
Total sales: 55000
Percentage increase: 80.0 %
```

5. **Average** fuel efficiency and percentage improvement

import numpy as np

fuel_efficiency = np.array([25, 30, 35, 28, 32])

average_efficiency = np.mean(fuel_efficiency)

model1 = fuel_efficiency[0]

model2 = fuel_efficiency[1]

percentage_improvement = ((model2 - model1) / model1) * 100

print("Average fuel efficiency:", average_efficiency

print("Percentage improvement:", percentage_improvement, "%")

Output:

```
# 5. Average fuel efficiency and percentage improvement

import numpy as np

fuel_efficiency = np.array([25, 30, 35, 28, 32])

average_efficiency = np.mean(fuel_efficiency)

model1 = fuel_efficiency[0]
model2 = fuel_efficiency[1]

percentage_improvement = ((model2 - model1) / model1) * 100

print("Average fuel efficiency:", average_efficiency)
print("Percentage improvement:", percentage_improvement, "%")
```

```
Average fuel efficiency: 30.0
Percentage improvement: 20.0 %
```

6. Total cost including discount and tax

prices = [100, 200, 150, 50]

quantities = [2, 1, 3, 2]

discount_rate = 10

tax_rate = 5

subtotal = sum(p * q for p, q in zip(prices, quantities))

discount = subtotal * discount_rate / 100

after_discount = subtotal - discount

tax = after_discount * tax_rate / 100

total_cost = after_discount + tax

print("Subtotal:", subtotal)

print("Discount:", discount)

print("Tax:", tax)

print("Total cost:", total_cost)

Output:

```
# 6. Total cost including discount and tax

prices = [100, 200, 150, 50]
quantities = [2, 1, 3, 2]

discount_rate = 10
tax_rate = 5

subtotal = sum(p * q for p, q in zip(prices, quantities))

discount = subtotal * discount_rate / 100
after_discount = subtotal - discount

tax = after_discount * tax_rate / 100
total_cost = after_discount + tax

print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax:", tax)
print("Total cost:", total_cost)
```

```
Subtotal: 950
Discount: 95.0
Tax: 42.75
Total cost: 897.75
```